

Amazon Brazil Data Analysis Project

Executive Summary

The project analyses Amazon Brazil's e-commerce data to identify trends, customer behaviours and preferences that could be leveraged in the Indian market. By examining various aspects like payment preferences, product pricing, customer segmentation, and seasonal patterns, this analysis aims to provide actionable insights for Amazon India's strategic decision-making.

Project Background

Amazon, as a global e-commerce leader, has achieved remarkable success across markets including the U.S., Europe, and Asia. In Brazil, Amazon has established itself as a key player connecting small and medium businesses with millions of customers. Given the similarities between Brazil and India—both featuring large populations and diverse consumer bases—there's a significant opportunity to apply successful strategies from the Brazilian market to India.

Objectives

1. Analyze customer demographics and behavior patterns
2. Identify regional trends and customer density
3. Track order lifecycles and understand product preferences
4. Evaluate seller performance and payment preferences
5. Develop actionable recommendations based on comprehensive data analysis

Methodology

The analysis employs PostgreSQL queries of varying complexity, from basic aggregations to advanced concepts including:

- Joins across multiple tables
- Common Table Expressions (CTEs)
- Window functions
- Subqueries and recursive queries
- Statistical calculations and categorization

Data Overview

The analysis utilizes seven interconnected tables:

- **Customers:** Customer information and demographics
- **Orders:** Order details including status and timestamps
- **Order Items:** Individual items with price and shipping information
- **Product:** Product specifications and categories
- **Seller:** Seller information and location
- **Payments:** Transaction details and payment methods

Through systematic analysis of these datasets, this project aims to uncover valuable insights that can inform Amazon India's market strategies and enhance customer experience.

DATA CLEANING

```
SELECT * FROM amazon_brazil.payments
WHERE payment_type = 'not_defined' ;
```

	order_id character varying (100)	payment_sequential integer	payment_type character varying (100)	payment_installments integer	payment_value numeric (10,2)
1	4637ca194b6387e2d538dc89b124b0ee	1	not_defined	1	0.00
2	00b1cb0320190ca0daa2c88b35206009	1	not_defined	1	0.00
3	c8c528189310eaa44a745b8d9d26908b	1	not_defined	1	0.00

During the data cleaning process, a set of transactions was identified with the payment_type labeled as 'not_defined' and payment_value = 0.00. These records are likely the result of placeholder values, data entry errors, or missing payment information.

Key Observations:

- All such records had a payment value of zero, which does not contribute to revenue.
- The payment_type was undefined, offering no insight into customer payment preferences or behavior.
- These records accounted for a very small fraction of the dataset.

Action Taken:

```
DELETE FROM amazon_brazil.payments
WHERE payment_type = 'not_defined' AND payment_value = 0.00;
```

Impact:

- Improved the accuracy of payment-type distribution analysis.
- Prevented misleading results when calculating average payment values or order proportions by payment method.

- Ensured that all records included in the analysis reflect valid transactions.

Analysis - I

1. To simplify its financial reports, Amazon India needs to standardize payment values. Round the average payment values to integer (no decimal) for each payment type and display the results sorted in ascending order.

```
SELECT payment_type,ROUND(AVG(payment_value)) AS rounded_avg_payment
FROM amazon_brazil.payments
GROUP BY payment_type
ORDER BY rounded_avg_payment ASC;
```

	payment_type character varying (150) 🔒	rounded_avg_payment numeric 🔒
1	voucher	66
2	debit_card	143
3	boleto	145
4	credit_card	163

Insights:

- **Voucher** has the lowest average payment (66), indicating it's used for smaller transactions.
- **Debit card** (143) and **boleto** (145) are similar in value, suggesting they may be used for mid-range transactions.
- **Credit card** has the highest average payment (163), likely indicating it's preferred for larger purchases.

Recommendations:

- **Voucher**: Promote more vouchers for smaller purchases, offering incentives to increase usage.

- **Debit card & Boleto:** Consider offering targeted promotions or discounts to encourage customers to spend more with these methods.
- **Credit card:** Focus on loyalty programs for higher-value transactions to encourage repeat spending.
- **Payment Diversification:** Analyze usage patterns and introduce new options if certain payment types are underutilized.

2.To refine its payment strategy, Amazon India wants to know the distribution of orders by payment type. Calculate the percentage of total orders for each payment type, rounded to one decimal place, and display them in descending order

```
SELECT payment_type, ROUND(COUNT(DISTINCT order_id) * 100.0 / (SELECT COUNT(DISTINCT order_id)
FROM amazon_brazil.payments), 1) AS percentage_orders
FROM amazon_brazil.payments GROUP BY payment_type ORDER BY percentage_orders DESC;
```

	payment_type character varying (150) 🔒	percentage_orders numeric 🔒
1	credit_card	76.9
2	boleto	19.9
3	voucher	3.9
4	debit_card	1.5

Insights:

- **Credit Card** dominates with 76.9% of total orders, suggesting it's the preferred payment method.
- **Boleto** follows with 19.9%, indicating it's also a significant choice.
- **Voucher** and **Debit Card** have smaller shares, 3.9% and 1.5% respectively, showing limited use compared to other methods.

Recommendations:

- **Credit Card:** Amazon India could explore ways to incentivize and maintain high usage of credit card payments.
- **Boleto:** It could be beneficial to offer promotions or loyalty programs to increase boleto usage.

- **Voucher & Debit Card:** Focus on strategies to boost these payment methods, such as offering special deals or discounts for users preferring them.
- **Payment Method Optimization:** Reassess payment options to ensure that customers have diverse and convenient payment choices.

3. Amazon India seeks to create targeted promotions for products within specific price range. Identify all products priced between 100 and 500 BRL that contain the word 'Smart' in their name. Display these products, sorted by price in descending order.

```
SELECT DISTINCT oi.product_id, price
FROM amazon_brazil.product p JOIN amazon_brazil.order_items oi ON p.product_id = oi.product_id
WHERE p.product_category_name LIKE '%smart%' AND oi.price BETWEEN 100 AND 500
ORDER BY oi.price DESC;
```

	product_id character varying (150)	price numeric (10,2)
1	1df1a2df8ad2b9d3aa49fd851e314...	439.99
2	7debe59b10825e89c1cbcc8b190c...	349.99
3	ca86b9fe16e12de698c955aedff0ae...	349.00
4	0e52955ca8143bd179b311ce454a...	335.00
5	7aeaa8f3e592e380c420e8910a717...	329.90
6	d1b571cd58267d8cac8b2afd6e288...	299.90
7	66ffe28d0fd53808d0535eee4b90a...	254.00
8	f06796447de379a26dde5fcac6a1a...	239.90
9	d3d5a1d52abe9a7d234908d873fc...	229.90
10	06ae026e430189633c2fbd0288c86...	217.36
11	49ef750dc5bf23e3788d4f614bc6d...	198.00
12	33bb7da523efcdef6cd2996cbf72d...	148.00
13	6f5795735ab2c629b22669fe889b7...	129.90
14	3626035966a7aee90d68108caeb...	124.90
15	aeaba104830f91586dae1bff90f54a...	123.90
16	630c84b1ce83ae0e9ddc05a14103...	110.00
17	3168b2696b15ca440b92afa9e011a...	109.90
18	dbd55362ec13c706503b1c71a506...	102.00
19	aeaba104830f91586dae1bff90f54a...	100.00

Insights:

1. **Product Demand:** The "Smart" category is popular within the **100 to 500 BRL range**, indicating strong interest in smart technology products at a mid-range price point.
2. **Price Distribution:** Higher-priced products (closer to 500 BRL) are prominent in the results, suggesting that customers are willing to spend more on smart devices within this range.
3. **Market Focus:** The data suggests that targeting the **"Smart"** category could be highly effective in promotions, as these products are consistently in the middle price range.

Recommendations:

1. **Focus on Premium Products:** Highlight **premium smart devices** priced near 500 BRL for high-end targeted promotions.

2. **Segmented Promotions:** Run **promotions and discounts** for products priced around 300-500 BRL, as they appear to have the most demand.
3. **Targeted Campaigns:** Use the **smart technology trend** to create **seasonal or flash sale campaigns** to boost sales in this category.
4. **Explore Loyalty Programs:** For the lower-priced products, consider introducing **loyalty rewards or bundle offers** to increase repeat purchases.

4. To identify seasonal sales patterns, Amazon India needs to focus on the most successful months. Determine the top 3 months with the highest total sales value, rounded to the nearest integer.

```
SELECT TO_CHAR(order_purchase_timestamp, 'YYYY-MM') AS order_month,  
SUM(payment_value) AS total_sales  
FROM amazon_brazil.orders o  
JOIN amazon_brazil.payments p ON o.order_id = p.order_id  
GROUP BY TO_CHAR(order_purchase_timestamp, 'YYYY-MM')  
ORDER BY total_sales DESC  
LIMIT 3;
```

	order_month text	total_sales numeric
1	2017-11	1194882.80
2	2018-04	1160785.48
3	2018-03	1159652.12

Insights:

1. **Top Sales Months:** The highest sales occurred in **November 2017**, followed by **April 2018** and **March 2018**, indicating these months have been the most successful.
2. **Seasonal Peaks:** **November 2017** saw the **highest total sales** (1,194,882.80 BRL), likely benefiting from holiday shopping or promotions.
3. **Consistency Across Months:** **March** and **April 2018** also show significant sales, suggesting **early spring** could be another high-sales period.

Recommendations:

1. **Focus on Strong Months:** Run **special promotions** in **November, March, and April** to maximize sales during historically high periods.

2. **Holiday and Seasonal Campaigns:** Take advantage of **November sales** with targeted **holiday campaigns** or early-bird promotions.
3. **Understand Sales Drivers:** Investigate specific events in **March and April** to identify what drove sales, helping to replicate this success in future months.

5. Amazon India is interested in product categories with significant price variations. Find categories where the difference between the maximum and minimum product prices is greater than 500 BRL.

```
SELECT distinct product_category_name, MAX(price) - MIN(price) AS price_difference
FROM amazon_brazil.product p JOIN amazon_brazil.order_items oi ON p.product_id = oi.product_id
GROUP BY product_category_name HAVING MAX(price) - MIN(price) > 500
ORDER BY price_difference DESC;
```

	product_category_name character varying (150)	price_difference numeric
1	utilidades_domesticas	6731.94
2	pcs	6694.50
3	artes	6495.50
4	eletroportateis	4792.50
5	instrumentos_musicais	4394.97
6	consoles_games	4094.81
7	esporte_lazer	4054.50
8	relogios_presentes	3990.91
9	[null]	3977.00
10	ferramentas_jardim	3923.65
11	bebes	3895.46
12	informatica_acessorios	3696.09
13	beleza_saude	3122.80
14	cool_stuff	3102.99
15	construcao_ferramentas_seguranca	3091.00
16	industria_comercio_e_negocios	3061.10
17	agro_industria_e_comercio	2977.01
18	portateis_casa_forno_e_cafe	2888.81
19	pet_shop	2495.10
20	eletronicos	2466.51
21	telefonica	2423.00

22	eletrodomesticos_2	2336.10
23	construcao_ferramentas_construcao	2299.15
24	automotivo	2254.51
25	eletrodomesticos	2083.81
26	cama_mesa_banho	1992.99
27	market_place	1954.01
28	moveis_decoracao	1894.10
29	construcao_ferramentas_ferramentas	1892.20
30	telefonica_fixa	1784.00
31	brinquedos	1695.09
32	fashion_bolsas_e_acessorios	1693.99
33	papelaria	1690.71
34	climatizacao	1588.10
35	smart	1444.50
36	dvds_blu_ray	1411.10
37	construcao_ferramentas_jardim	1341.08
38	moveis_cozinha_area_de_servico_jantar_e_jar...	1310.40
39	construcao_ferramentas_iluminacao	1277.49
40	malas_acessorios	1184.57
41	moveis_escritorio	1164.90
42	musica	1162.12

43	casa_construcao	1089.02
44	portateis_cozinha_e_preparadores_de_aliment...	1081.58
45	livros_interesse_geral	893.90
46	tablets_impressao_imagem	875.09
47	cine_foto	867.19
48	moveis_sala	826.09
49	casa_conforto	792.01
50	sinalizacao_e_seguranca	735.50
51	livros_importados	730.01
52	alimentos_bebidas	693.40
53	perfumaria	684.91
54	moveis_quarto	643.10
55	bebidas	617.00
56	audio	584.09
57	artigos_de_festas	563.21

Insights:

1. **Extremely High Price Variation:** Categories like "artes" (6495.50 BRL), "pcs" (6694.50 BRL), and "utilidades_domesticas" (6731.94 BRL) show substantial price differences. These likely include both basic and premium products, possibly with significant technological or brand-based differences.
2. **Wide Category Spread:** Over 50 product categories show price variation >500 BRL, suggesting broad product diversity across the catalog.
3. **Opportunity to Refine Pricing Strategy:** Some consumer segments, like "perfumaria" (684.91) or "smart" (1444.50), show moderate differences, indicating potential for tiered branding or bundling strategies.

Recommendations:

1. **Detailed Price Band Analysis:** Break down price ranges within top-variation categories to design **segment-specific offers** (e.g., basic, mid-tier, premium).
2. **Targeted Category Campaigns:** Use insights to launch **category-specific promotions**, highlighting **value products** and **premium alternatives** in the same space.
3. **UX Enhancements:** Provide filters and comparison tools in these categories to help users navigate wide price spreads and improve **decision-making**.

6. To enhance the customer experience, Amazon India wants to find which payment types have the most consistent transaction amounts. Identify the payment types with the least variance in transaction amounts, sorting by the smallest standard deviation first.

```
SELECT payment_type, STDDEV(payment_value) AS std_deviation
FROM amazon_brazil.payments
GROUP BY payment_type
ORDER BY std_deviation ASC;
```

	payment_type character varying (150) 🔒	std_deviation numeric 🔒
1	voucher	115.519185430458
2	boleto	213.581061475527
3	credit_card	222.119310741208
4	debit_card	245.793401040647

Insights:

1. Voucher payments show the lowest standard deviation (115.52), making them the most consistent in terms of transaction values.
2. Boleto (213.58) and Credit Card (222.12) payments exhibit moderate consistency, with only slight variability in transaction amounts.
3. Debit Card payments have the highest standard deviation (245.79), indicating greater inconsistency and wider variation in payment values compared to other methods.

Recommendations:

1. Promote voucher payments for categories or campaigns that require stable and predictable pricing.
2. Monitor debit card usage to identify customer segments or purchase patterns causing the high variability.
3. Highlight consistent payment methods (voucher, boleto) at checkout to improve user confidence and enhance the purchasing experience.

7. Amazon India wants to identify products that may have incomplete name in order to fix it from their end. Retrieve the list of products where the product category name is missing or contains only a single character.

```
SELECT product_id, product_category_name
FROM amazon_brazil.product
WHERE product_category_name IS NULL OR LENGTH(product_category_name) = 1;
```

	product_id [PK] character varying (150)	product_category_name character varying (150)
1	a41e356c76fab66334f36de622ecbd...	[null]
2	d8dee61c2034d6d075997acef1870...	[null]
3	56139431d72cd51f19eb9f7dae4d16...	[null]
4	46b48281eb6d663ced748f324108c...	[null]
5	5fb61f482620cb672f5e586bb132ea...	[null]
6	e10758160da97891c2fdcbc35f0f03...	[null]
7	39e3b9b12cd0bf8ee681bbc1c130fe...	[null]
8	794de06c32a626a5692ff50e4985d3...	[null]
9	7af3e2da474486a3519b0cba9dea8...	[null]
10	629beb8e7317703dcc5f35b5463fd2...	[null]
11	3a78f64aac654298e4b9aff32fc21818	[null]
12	bc815bba008d89458e428078c0b9...	[null]
13	6b82874c6b51b92913dcdb364eaaa...	[null]
Total rows: 614		Query complete 00:00:00.243

Insights:

- 614 products have missing or very short category names (length = 1), indicating potential data quality issues.

Recommendations:

1. Clean data by identifying and correcting these entries.
2. Label products with missing categories as "Needs Review."
3. Investigate the root cause and improve data entry validation.
4. Assess the business impact and prioritize high-impact fixes.

Analysis - II

1. Amazon India wants to understand which payment types are most popular across different order value segments (e.g., low, medium, high). Segment order values into three ranges: orders less than 200 BRL, between 200 and 1000 BRL, and over 1000 BRL. Calculate the count of each payment type within these ranges and display the results in descending order of count

```
SELECT
CASE
    WHEN payment_value < 200 THEN 'Low (<200 BRL)'
    WHEN payment_value BETWEEN 200 AND 1000 THEN 'Medium (200-1000 BRL)'
    ELSE 'High (>1000 BRL)'
END AS order_value_segment, payment_type, COUNT(*) AS count
FROM amazon_brazil.payments
GROUP BY order_value_segment, payment_type
ORDER BY count DESC;
```

	order_value_segment text	payment_type character varying (150)	count bigint
1	Low (<200 BRL)	credit_card	60548
2	Low (<200 BRL)	boleto	16444
3	Medium (200-1000 B...	credit_card	15303
4	Low (<200 BRL)	voucher	5476
5	Medium (200-1000 B...	boleto	3162
6	Low (<200 BRL)	debit_card	1287
7	High (>1000 BRL)	credit_card	944
8	Medium (200-1000 B...	voucher	286
9	Medium (200-1000 B...	debit_card	227
10	High (>1000 BRL)	boleto	178
11	High (>1000 BRL)	debit_card	15
12	High (>1000 BRL)	voucher	13
Total rows: 12 Query complete 00:00:00.129			

Insights:

1. The **credit card** payment type is the most popular across all segments, with a significant dominance in the "Low" and "Medium" order value segments.

2. **Boleto** is the second most popular payment method, particularly in the "Low" and "Medium" segments.
3. **Voucher** and **debit card** are less popular across all segments, with very few transactions in the "High" order value segment.
4. As order values increase, **credit card** remains the most frequent payment method, while **boleto** and **voucher** see a decrease in usage.

Recommendations:

1. Focus marketing and promotions around **credit card** and **boleto** payment options, particularly for low- to medium-value orders.
2. Consider offering incentives to encourage **voucher** and **debit card** usage, especially in higher-value orders.
3. Analyze the reasons behind the lower adoption of **voucher** and **debit card** to target potential barriers to usage in higher-value segments.

2. Amazon India wants to analyse the price range and average price for each product category. Calculate the minimum, maximum, and average price for each category, and list them in descending order by the average price.

```
SELECT product_category_name, MIN(price) AS min_price, MAX(price) AS max_price, AVG(price) AS avg_price
FROM amazon_brazil.product p
JOIN amazon_brazil.order_items oi ON p.product_id = oi.product_id
GROUP BY product_category_name
ORDER BY avg_price DESC;
```

	product_category_name character varying (150)	min_price numeric	max_price numeric	avg_price numeric
1	pcs	34.50	6729.00	1098.3405418719211823
2	portateis_casa_forno_e_cafe	10.19	2899.00	624.2856578947368421
3	eletrodomesticos_2	13.90	2350.00	476.1249579831932773
4	agro_industria_e_comercio	12.99	2990.00	341.6610426540284360
5	instrumentos_musicais	4.90	4399.87	281.6160000000000000
6	eletroportateis	6.50	4799.00	280.7784683357879234
7	portateis_cozinha_e_preparadores_de_aliment...	17.42	1099.00	264.5686666666666667
8	telefonos_fixos	6.00	1790.00	225.6931818181818182
9	construcao_ferramentas_seguranca	8.90	3099.90	208.9923711340206186
10	relogios_presentes	8.99	3999.90	200.9118770875083500
11	climatizacao	10.90	1599.00	185.2692255892255892
12	moveis_quarto	6.90	650.00	183.7502752293577982
13	pc_gamer	129.99	239.00	171.7722222222222222
14	cool_stuff	7.00	3109.99	167.3579689146469968
15	moveis_cozinha_area_de_servico_jantar_e_jar...	9.60	1320.00	164.8696441281138790
16	moveis_escritorio	25.00	1189.90	162.0110585452395033
17	musica	3.85	1165.97	158.7986842105263158
18	smart	15.50	1460.00	157.9316326530612245
19	construcao_ferramentas_construcao	0.85	2300.00	156.1271243243243243
20	construcao_ferramentas_ferramentas	6.80	1899.00	154.4072815533980583
21	industria_comercio_e_negocios	27.90	3089.00	148.0209328358208955

22	la_cuisine	24.00	389.00	146.7850000000000000
23	seguros_e_servicos	100.00	183.29	141.6450000000000000
24	automotivo	3.49	2258.00	139.9114572508266415
25	audio	14.90	598.99	139.2541208791208791
26	consoles_games	5.18	4099.99	138.4918381706244503
27	casa_construcao	8.97	1097.99	137.5631125827814570
28	moveis_sala	13.90	839.99	137.0110536779324056
29	construcao_ferramentas_illuminacao	12.51	1290.00	135.1315789473684211
30	casa_conforto	6.99	799.00	134.9586175115207373
31	bebes	3.54	3899.00	134.3998987920339536
32	beleza_saude	1.20	3124.00	130.1868495135582695
33	malas_acessorios	13.33	1197.90	128.5790642201834862
34	brinquedos	4.90	1699.99	117.5090206561360875
35	perfumaria	4.99	689.90	116.7373120795554256
36	informatica_acessorios	3.90	3699.99	116.6096901011653221
37	artes	3.50	6499.00	115.8021052631578947
38	moveis_colchao_e_estofado	44.70	399.99	114.9494736842105263
39	esporte_lazer	4.50	4059.00	114.2520686058639472
40	[null]	3.00	3980.00	111.9995508421709295
41	c	104.90	114.90	111.7750000000000000
42	ferramentas_jardim	6.35	3930.00	111.1497649227932703

43	pet_shop	2.90	2498.00	110.14650000000000
44	sinalizacao_e_seguranca	4.50	740.00	108.0865829145728643
45	construcao_ferramentas_jardim	8.82	1349.90	108.0499579831932773
46	artigos_de_festas	7.99	571.20	104.3065116279069767
47	eletrodomesticos	6.19	2090.00	102.2119346405228758
48	cine_foto	6.80	873.99	96.29805555555555
49	dvds_blu_ray	8.90	1420.00	93.7404687500000000
50	cama_mesa_banho	6.99	1999.98	93.2744727568446734
51	papelaria	2.29	1693.00	91.7643998410174881
52	market_place	9.99	1964.00	91.2490996784565916
53	utilidades_domesticas	3.06	6735.00	90.8057054597701149
54	tablets_impressao_imagem	14.90	889.99	90.7037349397590361
55	fashion_calcados	25.90	365.00	90.18450000000000
56	moveis_decoracao	4.90	1899.00	87.5645735223450264
57	livros_interesse_geral	6.00	899.90	84.7321518987341772
58	fashion_roupa_masculina	14.90	292.50	81.80166666666666
59	livros_importados	18.99	749.00	77.2516949152542373
60	artes_e_artesanato	9.80	289.49	75.58375000000000
61	fashion_bolsas_e_acessorios	6.00	1699.99	75.2454652880354505
62	fashion_underwear_e_moda_praia	3.90	192.00	72.8362595419847328
63	livros_tecnicos	9.95	384.93	71.5208239700374532

64	fashion_roupa_infanto_juvenil	39.99	110.00	71.23125000000000
65	telefonos	5.00	2428.00	71.2197955145118734
66	fashion_esporte	24.99	225.00	70.65033333333333
67	t	68.90	68.90	68.90000000000000
68	w	59.90	59.90	59.90000000000000
69	bebidas	6.00	623.00	59.178627968377309
70	fashion_roupa_feminina	19.90	198.90	58.40916666666666
71	eletronicos	3.99	2470.50	57.9283586406362979
72	alimentos	9.99	274.99	57.6341372549019608
73	artigos_de_natal	5.50	295.00	57.5216993464052288
74	alimentos_bebidas	6.50	699.90	54.6024460431654676
75	cds_dvds_musicais	45.00	65.00	52.1428571428571429
76	fraldas_higiene	25.00	139.89	40.1946153846153846
77	flores	15.60	65.90	33.63757575757576
78	casa_conforto_2	12.90	219.99	25.34233333333333
79	f	21.70	21.70	21.70000000000000

Insights:

1. The "**pcs**" category has the highest average price (1098.34 BRL), with a wide price range between 34.50 BRL (minimum) and 6729.00 BRL (maximum).
2. "**portateis_casa_forno_e_cafe**" and "**eletrodomesticos_2**" are other categories with relatively high average prices, at 624.29 BRL and 476.12 BRL respectively.
3. Categories like "**f**", "**c**", and "**t**" have the lowest average prices (21.70 BRL, 59.90 BRL, and 68.90 BRL), with narrow price ranges.
4. "**bebes**" and "**beleza_saude**" also have low average prices, at 134.40 BRL and 130.19 BRL, respectively, but with broader price ranges compared to categories with very low average prices.

Recommendations:

1. Focus on premium products within categories like "**pcs**" and "**portateis_casa_forno_e_cafe**" to target high-value customers.
2. Explore potential marketing strategies for "**f**", "**c**", and "**t**" to increase customer engagement despite their low average prices.
3. Consider bundling products in categories with low average prices, like "**bebes**", to increase perceived value.

3. Amazon India wants to identify the customers who have placed multiple orders over time. Find all customers with more than one order, and display their customer unique IDs along with the total number of orders they have placed.

	customer_id character varying (150)	total_orders bigint
1	a86b68dd9da561c8b4cf830e2f0f47...	16
2	827d7240a35889668129a7b10aa00...	16
3	e28c7984cdad31abb88f1e19cd862...	16
4	a8e5ab9fae7e21ef7e843734f55602...	16
5	751e579ed6c435ddbe018ccc5b68e...	16
6	64e37599865347bc3329d0d79faa8...	16
7	b5dff22ad6f2531a66adde7810935f...	16
8	af046fdbbda4a307d96801ebe6148...	16
9	dd24965dd5aa551df36ce4baf1804...	16
10	120cfe05591775d14f4852a9080a2...	16
11	32a079c19bb9adf352e0e93c4fb9bd...	16
12	c4bd3aefb216827f4698bd9d2b166...	16
13	3922e40a232b96383899bb8b6f111...	16
14	3249a3fb2ccd4b5f7103a6e759c0d8...	16
15	7fe3893a7bf9572f35028d2d3ef9f85d	16
16	cd3ce27aac0c6ecc4a3ca29a93f07c...	16
17	ef6fffb44c4dd02c0e35d6b6f9acf31e	16
18	1d5c5823a964c7891fca921536fb24...	16
19	ba5a299fef88320dbdadd124f5333b...	16
Total rows: 215		Query complete 00:00:00.094

```
Select customer_id, count(*) as total_orders
from amazon_brazil.orders
group by customer_id
having count(*) > 1
order by total_orders desc;
```

Insights:

1. **Frequent Orders:** A few customers placed 16 orders, showing high loyalty. Others have fewer orders, indicating potential for increased engagement.
2. **Retention Opportunities:** Customers with fewer orders (4-6) suggest room for improvement in retention.
3. **Top Customers:** Focus on those with the most orders for loyalty programs.

Recommendations:

1. **Loyalty Programs:** Reward high-frequency customers with exclusive deals.
2. **Retention Campaigns:** Engage customers with 4-6 orders through personalized offers and reminders.
3. **Customer Segmentation:** Tailor marketing strategies for high, moderate, and low-frequency buyers.
4. **Targeted Ads:** Focus on products bought by frequent customers for more precise marketing.
5. **Referral Programs:** Encourage loyal customers to refer others for rewards.

4. Amazon India wants to categorize customers into different types ('New – order qty. = 1'; 'Returning' –order qty. 2 to 4; 'Loyal' – order qty. >4) based on their purchase history. Use a temporary table to define these categories and join it with the customers table to update and display the customer types.

```
CREATE TEMPORARY TABLE customer_types AS
SELECT o.customer_id,
CASE
    WHEN COUNT(o.order_id) = 1 THEN 'New'
    WHEN COUNT(o.order_id) BETWEEN 2 AND 4 THEN 'Returning'
    WHEN COUNT(o.order_id) > 4 THEN 'Loyal'
END AS customer_type
FROM amazon_brazil.orders o
GROUP BY o.customer_id;
SELECT customer_id, customer_type
FROM customer_types ;
```

```
SELECT CUSTOMER_TYPE, COUNT(*) AS COUNT
FROM CUSTOMER_TYPES
GROUP BY CUSTOMER_TYPE ;
```

	customer_id character varying (150)	customer_type text
1	00012a2ce6f8dcda20d059ce98491...	New
2	000161a058600d5901f007fab4c271...	New
3	0001fd6190edaaf884bcfa3d49edf079	New
4	0002414f95344307404f0ace7a26f1...	New
5	000379cdec625522490c315e70c7a...	New
6	0004164d20a9e969af783496f34086...	New
7	000419c5494106c306a97b5635748...	New
8	00046a560d407e99b969756e0b10f...	New
9	00050bf6e01e69d5c0fd612f1bcfb69c	New
10	000598caf2ef4117407665ac332751...	New
11	00062b33cb9f6fe976afdcff967ea74d	New
Total rows: 98348		Query complete 00:00:00.723

	customer_type text	count bigint
1	New	98133
2	Returning	117
3	Loyal	98

Insights:

1. **Majority Are New:** Most customers (98,133) have made only one purchase, indicating a high drop-off after the first order.
2. **Loyal Customers Are Few:** Only 98 customers fall into the 'Loyal' category, showing a significant opportunity to boost retention.
3. **Moderate Engagement:** 117 customers are 'Returning' (2–4 orders), suggesting a small but growing group that may convert to loyal with the right push.

Recommendations:

1. **Improve First Experience:** Enhance post-purchase engagement to convert 'New' customers to 'Returning'.
2. **Re-engagement Campaigns:** Target 'New' customers with timely offers or reminders.
3. **Upsell to 'Returning':** Encourage this group with loyalty points or exclusive discounts to promote more frequent purchases.
4. **Reward Loyalty:** Strengthen relationships with 'Loyal' customers through premium rewards, early access, or recognition programs.

5. Amazon India wants to know which product categories generate the most revenue. Use joins between the tables to calculate the total revenue for each product category. Display the top 5 categories.

```
SELECT p.product_category_name, SUM(oi.price) AS total_revenue
FROM amazon_brazil.product p JOIN amazon_brazil.order_items oi ON p.product_id = oi.product_id
GROUP BY p.product_category_name
ORDER BY total_revenue DESC
LIMIT 5;
```

	product_category_name character varying (150)	total_revenue numeric
1	beleza_saude	1257865.34
2	relogios_presentes	1203060.32
3	cama_mesa_banho	1032268.59
4	esporte_lazer	985881.10
5	informatica_acessorios	910605.07

Insights:

- 1. **Top Category – Beauty & Health:** 'beleza_saude' leads in revenue, indicating high demand in personal care.
- 2. **Watches & Gifts Popular:** 'relogios_presentes' ranks second, showing strong performance in lifestyle/gifting.
- 3. **Home Essentials Thrive:** 'cama_mesa_banho' suggests consistent consumer investment in home products.
- 4. **Active Lifestyle Products Sell Well:** 'esporte_lazer' indicates interest in fitness and recreation.
- 5. **Tech Accessories Are Lucrative:** 'informatica_acessorios' reflects growing tech usage and gadget reliance.

Recommendations:

- 1. **Promote Top Sellers:** Increase visibility and ad spend on the top 3 categories.
- 2. **Bundle Products:** Create combo offers within high-performing categories to boost order value.
- 3. **Stock Strategically:** Ensure inventory is aligned with demand in these high-revenue areas.
- 4. **Cross-Sell Opportunities:** Suggest related items from top categories during checkout.

Analysis - III

1. The marketing team wants to compare the total sales between different seasons. Use a subquery to calculate total sales for each season (Spring, Summer, Autumn, Winter) based on order purchase dates, and display the results. Spring is in the months of March, April and May. Summer is from June to August and Autumn is between September and November and rest months are Winter.

```
SELECT season, SUM(price) AS total_sales
FROM
(SELECT oi.price,
CASE
WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN (3, 4, 5) THEN 'Spring'
WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN (6, 7, 8) THEN 'Summer'
WHEN EXTRACT(MONTH FROM o.order_purchase_timestamp) IN (9, 10, 11) THEN 'Autumn'
ELSE 'Winter'
END AS season
FROM amazon_brazil.orders o JOIN amazon_brazil.order_items oi ON o.order_id = oi.order_id) AS season_sales
GROUP BY season
ORDER BY total_sales desc;
```

	season text	total_sales numeric
1	Spring	4216721.54
2	Summer	4120359.62
3	Winter	2905750.03
4	Autumn	2348812.51

Insights:

1. **Spring Leads in Sales:** With over **4.2M** in revenue, Spring is the most profitable season.
2. **Summer is Close Behind:** Summer sales are also strong, suggesting good consumer activity during mid-year.
3. **Winter and Autumn Lag:** Lower sales in Winter and Autumn may indicate reduced shopping activity or fewer promotions.

Recommendations:

1. **Boost Autumn/Winter Sales:** Launch seasonal promotions or campaigns to increase traction during slower months.
2. **Leverage Spring Momentum:** Plan major product launches or marketing pushes during Spring.
3. **Analyse Seasonal Preferences:** Tailor offerings by season to match customer behaviour and needs.

2. The inventory team is interested in identifying products that have sales volumes above the overall average. Write a query that uses a subquery to filter products with a total quantity sold above the average quantity.

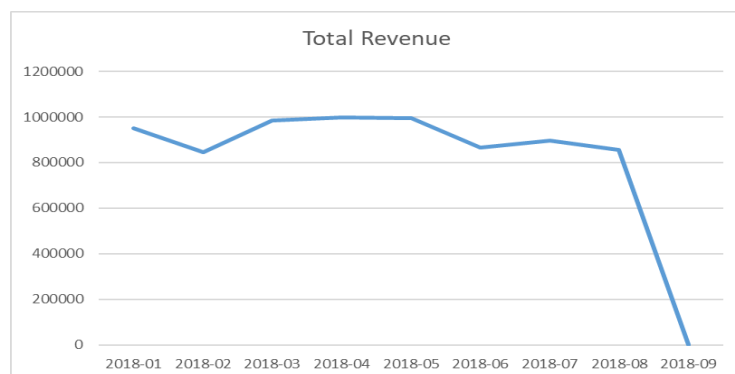
```
SELECT product_id, total_quantity_sold
FROM (SELECT product_id, COUNT(order_item_id) AS total_quantity_sold
FROM amazon_brazil.order_items
GROUP BY product_id) AS product_sales
WHERE total_quantity_sold > (SELECT AVG(total_quantity_sold)
FROM (SELECT COUNT(order_item_id) AS total_quantity_sold
FROM amazon_brazil.order_items
GROUP BY product_id) AS avg_sales);
```

	product_id character varying (150)	total_quantity_sold bigint
1	3d5837f86205fe83f03fb5f7e4d5b9cf	11
2	afeeea6271148ee1bb15173b8187c4...	53
3	434487f82b5c35646bd8155cf19461...	4
4	e5063ce7fff1cf7cd528dc4c1e7dcba8	4
5	b25a0f93e25104798df2d1664495d1...	4
6	6639a238ead6779d6ef0b3eea56f9f86	4
7	dceb3f67aef3484498a7caa4ba50f484	6
8	3c4e3782469a0f1ac459dc6c47ebef31	4
9	ecaaaccb5eb3102553f001d62db638...	6
10	98ad26989524a790f1d29686025b6f	4
Total rows: 6366		Query complete 00:00:00.975

3. To understand seasonal sales patterns, the finance team is analysing the monthly revenue trends over the past year (year 2018). Run a query to calculate total revenue generated each month and identify periods of peak and low sales. Export the data to Excel and create a graph to visually represent revenue changes across the months.

```
SELECT EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month, SUM(oi.price) AS total_revenue
FROM amazon_brazil.orders o
JOIN amazon_brazil.order_items oi ON o.order_id = oi.order_id
WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
GROUP BY month
ORDER BY month;
```

	month numeric	total_revenue numeric
1	1	950030.36
2	2	844178.71
3	3	983213.44
4	4	996647.75
5	5	996517.68
6	6	865124.31
7	7	895507.22
8	8	854686.33
9	9	145.00



Insights:

- Peak Sales:** The months with the highest sales were **April, May, and March**, with total revenues close to 1 million.
- Low Sales:** **February** and **September** had lower sales, with February being significantly lower. September, in particular, saw a sharp decline in revenue to **145.00**, which may indicate a data issue or an unexpected event.

3. **Seasonal Trends:** The months of **March through May** seem to show the highest performance, suggesting possible seasonal demand peaks in spring.

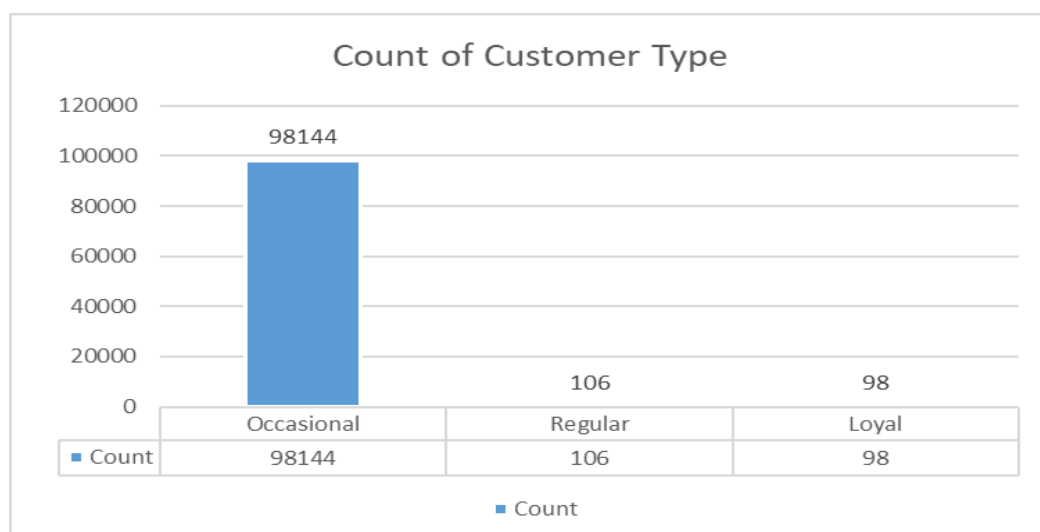
Recommendations:

1. Investigate the **September data anomaly** (145.00) to ensure accuracy.
2. Focus marketing and promotional efforts around the **spring months (March-May)** to capitalize on peak sales trends.
3. Consider **targeted campaigns** in **February** and **September**, given their relatively lower sales.
4. Explore whether external factors (e.g., holidays or events) contributed to the peaks and troughs.

4. A loyalty program is being designed for Amazon India. Create a segmentation based on purchase frequency: 'Occasional' for customers with 1-2 orders, 'Regular' for 3-5 orders and 'Loyal' for more than 5 orders. Use a CTE to classify customers and their count and generate a chart in Excel to show the proportion of each segment.

```
WITH CustomerSegmentation AS
(SELECT customer_id, COUNT(order_id) AS order_count,
CASE
    WHEN COUNT(order_id) <= 2 THEN 'Occasional'
    WHEN COUNT(order_id) BETWEEN 3 AND 5 THEN 'Regular'
    ELSE 'Loyal'
END AS customer_type
FROM amazon_brazil.orders
GROUP BY customer_id)
SELECT customer_type, COUNT(*) AS count FROM CustomerSegmentation
GROUP BY customer_type
ORDER BY count DESC;
```

	customer_type 	count 
1	Occasional	98144
2	Regular	106
3	Loyal	98



Insights:

1. **Occasional Customers:** The majority of customers fall into the "Occasional" category, with **98,144** customers making 1-2 orders.
2. **Regular Customers:** A relatively small group, **106** customers, make 3-5 orders.
3. **Loyal Customers:** An even smaller group, **98** customers, have placed more than 5 orders.

Recommendations:

1. **Focus on "Occasional" Segment:** Since most customers are occasional buyers, offer incentives (e.g., discounts, targeted promotions) to encourage them to become "Regular" or "Loyal" customers.
2. **Loyalty Program Tailoring:** Create exclusive rewards for "Loyal" customers to retain them and increase customer lifetime value.
3. **Promote Regular Purchases:** For the "Regular" segment, offer personalized deals or notifications to encourage them to become "Loyal."
4. **Proportion Chart:** In Excel, create a pie chart to visualize the distribution of each segment:
 - a. "Occasional" should dominate the chart.
 - b. "Regular" and "Loyal" segments will form smaller portions.

5. Amazon wants to identify high-value customers to target for an exclusive rewards program. You are required to rank customers based on their average order value to find the top 20 customers.

```
WITH order_totals AS (  
  SELECT  
    O.customer_id,  
    O.order_id,  
    SUM(oi.price) AS order_value  
  FROM amazon_brazil.orders O  
  JOIN amazon_brazil.order_items oi ON O.order_id = oi.order_id  
  GROUP BY O.customer_id, O.order_id  
)  
customer_avg_order AS (  
  SELECT  
    Customer_id,  
    AVG(order_value) AS avg_order_value  
  FROM order_totals  
  GROUP BY customer_id  
)  
SELECT  
  Customer_id,  
  avg_order_value,  
  RANK() OVER (ORDER BY avg_order_value DESC) AS customer_rank  
FROM customer_avg_order  
LIMIT 20;
```

	customer_id character varying (150)	avg_order_value numeric	customer_rank bigint
1	1617b1357756262bfa56ab541c47b...	13440.0000000000000000	1
2	ec5b2ba62e574342386871631fafd...	7160.0000000000000000	2
3	c6e2731c5b391845f6800c97401a4...	6735.0000000000000000	3
4	f48d464a0baaea338cb25f816991ab...	6729.0000000000000000	4
5	3fd6777bbce08a352fddd04e4a7cc8...	6499.0000000000000000	5
6	05455dfa7cd02f13d132aa7a6a972...	5934.6000000000000000	6
7	df55c14d1476a9a3467f131269c24...	4799.0000000000000000	7
8	24bbf5fd2f2e1b359ee7de94defc4a15	4690.0000000000000000	8
9	e0a2412720e9ea4f26c1ac985f6a73...	4599.9000000000000000	9
10	3d979689f636322c62418b6346b1c...	4590.0000000000000000	10
11	cc803a2c412833101651d3f90ca7d...	4400.0000000000000000	11
12	1afc82cd60e303ef09b4ef9837c950...	4399.8700000000000000	12
13	35a413c7ca3c69756cb75867d6311...	4099.9900000000000000	13
14	e9b0d0eb3015ef1c9ce6cf5b9dcbee...	4059.0000000000000000	14
15	c6695e3b1e48680db36b487419fb0...	3999.9000000000000000	15
16	926b6a6fb8b6081e00b335edaf578...	3999.0000000000000000	16
17	3be2c536886b2ea4668eced3a80dd...	3980.0000000000000000	17
18	31e83c01fce824d0ff786fcd48dad009	3930.0000000000000000	18
19	eb7a157e8da9c488cd4ddc48711f1...	3899.0000000000000000	19
20	19b32919fa1198aefc0773ee2e46e6...	3700.0000000000000000	20

Insights:

1. The top customer spends **13,440** on average, with the top 5 ranging from **13,440** to **6,499**.
2. The 20th customer has an average of **3,700**.

Recommendations:

1. Target the top 20 customers for an exclusive rewards program.
2. Analyse their spending patterns for personalized marketing.
3. Create a bar chart in Excel to visualize their ranks and order values.

6. Amazon wants to analyse sales growth trends for its key products over their lifecycle. Calculate monthly cumulative sales for each product from the date of its first sale. Use a recursive CTE to compute the cumulative sales for each product month by month.

```
WITH recursive product_sales AS (
    SELECT
        oi.product_id,
        DATE_TRUNC('month', o.order_purchase_timestamp) AS sale_month,
        SUM(oi.price) AS monthly_sales
    FROM amazon_brazil.order_items oi
    JOIN amazon_brazil.orders o ON oi.order_id = o.order_id
    GROUP BY oi.product_id, DATE_TRUNC('month', o.order_purchase_timestamp)
),
recursive_sales AS (
    SELECT ps.product_id, ps.sale_month, ps.monthly_sales, ps.monthly_sales AS total_sales
    FROM product_sales ps
    WHERE NOT EXISTS (
        SELECT 1 FROM product_sales ps2
        WHERE ps2.product_id = ps.product_id
        AND ps2.sale_month < ps.sale_month
    )
    UNION ALL
    SELECT ps.product_id, ps.sale_month, ps.monthly_sales,
        rs.total_sales + ps.monthly_sales AS total_sales
    FROM product_sales ps
    JOIN recursive_sales rs
    ON ps.product_id = rs.product_id
    AND ps.sale_month = rs.sale_month + INTERVAL '1 month'
)
SELECT product_id, sale_month, total_sales
FROM recursive_sales
ORDER BY product_id, sale_month;
```

	product_id character varying (150)	sale_month timestamp without time zone	total_sales numeric
1	00066f42aeeb9f3007548bb9d3f3...	2018-05-01 00:00:00	101.65
2	00088930e925c41fd95ebfe695fd2...	2017-12-01 00:00:00	129.90
3	0009406fd7479715e4bef61dd91f...	2017-12-01 00:00:00	229.00
4	000b8f95fcb9e009648827831776...	2018-08-01 00:00:00	117.80
5	000d9be29b5207b54e86aa1b1ac...	2018-04-01 00:00:00	199.00
6	0011c512eb256aa0dbbb544d8dff...	2017-12-01 00:00:00	52.00
7	00126f27c813603687e6ce486d90...	2017-09-01 00:00:00	498.00
8	001795ec6f1b187d37335e1c4704...	2017-10-01 00:00:00	38.90
9	001795ec6f1b187d37335e1c4704...	2017-11-01 00:00:00	116.70
10	001795ec6f1b187d37335e1c4704...	2017-12-01 00:00:00	350.10
Total rows: 43039		Query complete 00:00:14.829	

7. To understand how different payment methods affect monthly sales growth, Amazon wants to compute the total sales for each payment method and calculate the month-over-month growth rate for the past year (year 2018). Write query to first calculate total monthly sales for each payment method, then compute the percentage change from the previous month.

```
WITH monthly_sales AS (
  SELECT
    payment_type,
    TO_CHAR(o.order_purchase_timestamp, 'YYYY-MM') AS sale_month,
    SUM(p.payment_value) AS monthly_total
  FROM amazon_brazil.payments p
  JOIN amazon_brazil.orders o ON p.order_id = o.order_id
  WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) = 2018
  GROUP BY payment_type, TO_CHAR(o.order_purchase_timestamp, 'YYYY-MM')
),
sales_with_lag AS (
  SELECT
    payment_type,
    sale_month,
    monthly_total,
    LAG(monthly_total) OVER (PARTITION BY payment_type ORDER BY sale_month) AS prev_month_total
  FROM monthly_sales
)
SELECT
  payment_type,
  sale_month,
  monthly_total,
  ROUND(100.0 * (monthly_total - COALESCE(prev_month_total, 0)) / NULLIF(prev_month_total, 0), 2 )
  AS monthly_change
FROM sales_with_lag
ORDER BY payment_type, sale_month;
```

	payment_type character varying (150) 🔒	sale_month text 🔒	monthly_total numeric 🔒	monthly_change numeric 🔒
1	boleto	2018-01	204844.66	[null]
2	boleto	2018-02	183112.72	-10.61
3	boleto	2018-03	191538.02	4.60
4	boleto	2018-04	193547.09	1.05
5	boleto	2018-05	195378.93	0.95
6	boleto	2018-06	153350.28	-21.51
7	boleto	2018-07	198041.24	29.14
8	boleto	2018-08	143805.90	-27.39
9	credit_card	2018-01	868880.38	[null]
10	credit_card	2018-02	778803.00	-10.37
11	credit_card	2018-03	933770.10	19.90
12	credit_card	2018-04	934306.00	0.06
13	credit_card	2018-05	927556.35	-0.72
14	credit_card	2018-06	811508.56	-12.51
15	credit_card	2018-07	803674.49	-0.97
16	credit_card	2018-08	797648.89	-0.75
17	debit_card	2018-01	11543.55	[null]
18	debit_card	2018-02	7469.53	-35.29
19	debit_card	2018-03	8375.11	12.12
20	debit_card	2018-04	10782.53	28.74
21	debit_card	2018-05	9710.74	-9.94

22	debit_card	2018-06	35672.62	267.35
23	debit_card	2018-07	44866.18	25.77
24	debit_card	2018-08	46001.33	2.53
25	voucher	2018-01	29735.59	[null]
26	voucher	2018-02	23078.09	-22.39
27	voucher	2018-03	25968.89	12.53
28	voucher	2018-04	22149.86	-14.71
29	voucher	2018-05	21336.13	-3.67
30	voucher	2018-06	23349.04	9.43
31	voucher	2018-07	19958.84	-14.52
32	voucher	2018-08	34969.20	75.21
33	voucher	2018-09	4439.54	-87.30
34	voucher	2018-10	589.67	-86.72

Insights:

1. **Boleto** and **Credit Card** show fluctuating sales, with Boleto peaking in July (+29.14%) and Credit Card declining from March onwards.
2. **Debit Card** has strong growth, especially in June (+267.35%).
3. **Voucher** experiences sharp declines in Sept (-87.30%) and Oct (-86.72%).

Recommendations:

1. Boost promotions for **Boleto** and **Voucher** during low growth months.
2. Focus on growing **Debit Card** sales.
3. Investigate reasons behind **Credit Card** decline.
4. Visualize data in Excel for clearer trend analysis.